

The Editor-in-Chief
Journal of Modern Power Systems and Clean Energy

Dear Editor-in-Chief,

We submit the manuscript “*Revisiting Metaheuristic Comparisons for Continuous Wind Farm Layout Optimization: Baseline Configuration, Component Analysis and a Two-Phase PSO–VNS Design*” by Prince Solanki, Pulkit Dwivedi, Vanita Garg and Vinay Shukla for consideration as a regular paper in the *Journal of Modern Power Systems and Clean Energy*.

What the paper contributes. Many metaheuristics have been proposed for wind farm layout optimization, but their comparisons are hard to trust because baselines, budgets, feasibility and statistics are handled differently. We define a controlled evaluation protocol (equal budgets that count every objective and gradient call, seed-paired runs, feasibility-aware ranking, case-mean and run-level tests with Holm correction, an ablation with a random-sampling control, robustness re-evaluation, open code and data) and apply it to eight methods on 68 benchmark cases, a Horns Rev 1 block and IEA37 Case Study 1. Three findings follow. (M1) Baseline configuration changes the conclusions: the PSO setting $w = 0.7$, $c_1 = c_2 = 2$, common in layout studies, violates the particle convergence condition; with it PSO ranks fifth and salp-swarm hybrids appear better, whereas with the Clerc–Kennedy constriction setting PSO ranks second and outperforms every salp-swarm method. (M2) In two-phase swarm–VNS hybrids the VNS phase helps every swarm, but the salp-swarm phases are statistically tied with random sampling and the Laplace step adds nothing; only a PSO phase beats random sampling. (M3) The resulting simple design, PSO followed by basic VNS (PSO–VNS), is the most consistent performer and is feasible in every run. We state its limits plainly: it is not significantly better than PSO over all cases, VNS alone is best from feasible starts, PSO ranks first under a Gaussian wake re-evaluation, and on IEA37 a gap to the gradient-based published optima remains. The paper closes with recommendations for benchmarking layout methods.

Fit to the journal. The layout fixes the energy yield of a wind farm for its lifetime, and layout optimizers are used at the planning stage to compare sites, turbine counts and spacing rules. Such decisions can rely on an optimizer only if its reported advantage survives a fair comparison. The paper gives planners and method developers a reproducible way to evaluate layout methods, a simple reference method, and checks of the conclusions against the power-curve, wake-model and spacing assumptions. It continues the journal’s line of work on metaheuristic layout optimization.

Disclosure of an earlier submission. A related manuscript by the same authors, “*A Hybrid Laplacian Salp Swarm and Variable Neighborhood Search Algorithm for Continuous Wind Farm Layout Optimization*” (Manuscript ID [TBD: ID of the earlier MPCE submission]; decision: [TBD: e.g., rejected]), was previously submitted to the journal. The present manuscript is a new paper with a different thesis, and the 22 comments of the three reviewers shaped it directly. The requests for equal-budget comparisons with stronger baselines and proper statistics led us to recheck the PSO baseline, which revealed M1. The request for an ablation of the Laplace mechanism led to the component analysis with a random-sampling control, which revealed M2 and made us drop the salp-swarm phase. The requests for a real site and for tests of the power curve, wake model and 4D spacing led to the Horns Rev 1 and IEA37 cases and to the sensitivity analysis. A point-by-point mapping of all 22 comments is attached; it also names the points that are only partly addressed.

The manuscript is original, not published and not under consideration elsewhere; all authors approved its submission and declare no competing interests. Code and data: [TBD: Zenodo DOI].

Sincerely,

Prince Solanki [TBD: confirm corresponding author], on behalf of all authors
Department of Mathematics, School of Sciences, IILM University, Greater Noida, Uttar Pradesh,
India; prince@iilm.edu